

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 Total Marks:50 Annexure A

CONCEPT MAPPING

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability). Perform the following tasks: a.

- Show how each attack leads to specific threats and risks
- b. Map how security goals help mitigate them Provide one real-world example for each link
2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.
3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks: a. Show differences and similarities
- i. Map how each affects the Data structure and Security level
 - ii. Analyze which is more vulnerable to attacks and why
4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
- a. Show relationships between these concepts
 - b. Analyze how each contributes to secure communication
 - c. Include one real-world application for each
5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
- a. Show how each concept is mathematically connected
 - b. Map their role in encryption/decryption
 - c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

QUESTION 2 – CONCEPT MAP

Comparative Concept Map of Substitution Techniques

Topic

Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher

1. Introduction

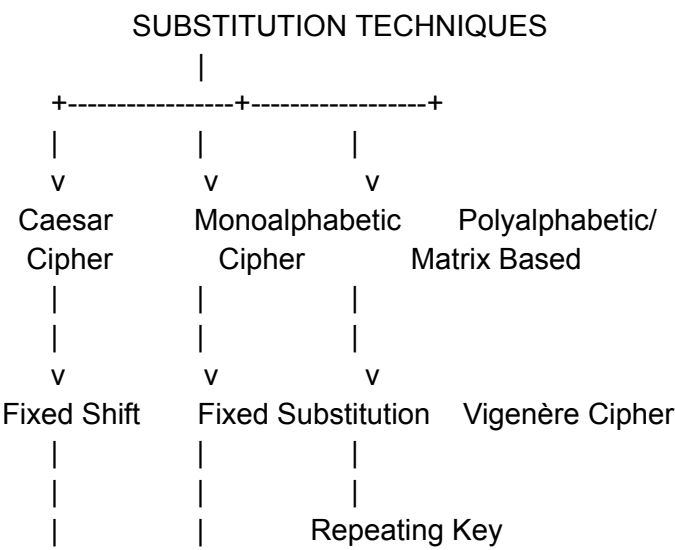
Substitution ciphers are classical encryption techniques in which plaintext symbols are replaced with other symbols according to a particular rule or key.

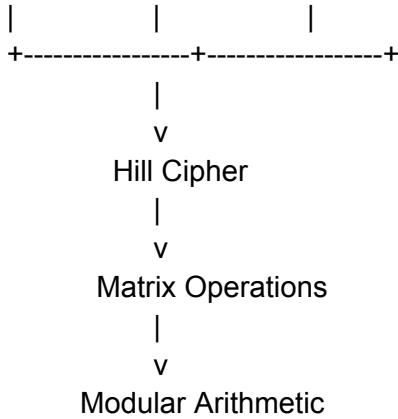
The four techniques considered are:

- 1. Caesar Cipher
- 2. Monoalphabetic Cipher
- 3. Vigenère Cipher
- 4. Hill Cipher

They differ in their **key usage, complexity, resistance to attacks, and security strength**.

2. Concept Map





3. Caesar Cipher

The Caesar cipher replaces every plaintext letter with another letter shifted by a fixed number of positions.

Formula

Encryption:

$$C = (P + K) \bmod 26$$

Decryption:

$$P = (C - K) \bmod 26$$

Where:

- P = plaintext value
- C = ciphertext value
- K = shift key

Example

Plaintext:

HELLO

Key:

3

Encryption:

H → K

E → H

L → O

L → O

O → R

Ciphertext:

KHOOR

Characteristics

- Key usage: Single numerical shift.
- Complexity: Very low.
- Security strength: Very weak.
- Key space: Only 26 possible shifts.

Weakness

An attacker can simply try all possible shifts using a **brute-force attack**.

4. Monoalphabetic Cipher

A monoalphabetic cipher uses a complete substitution alphabet.

Example:

Plain : ABCDEFGHIJKLMNOPQRSTUVWXYZ

Cipher: QWERTYUIOPASDFGHJKLZXCVBNM

Each plaintext character always maps to the same ciphertext character.

Example

HELLO

could become:

ITSSG

depending on the substitution alphabet.

Characteristics

- Key usage: Complete substitution alphabet.
- Complexity: Higher than Caesar.
- Security strength: Better than Caesar but still weak.
- Main weakness: Frequency analysis.

For example, if a particular ciphertext character occurs very frequently, an attacker may infer that it represents a common plaintext letter such as E.

5. Vigenère Cipher

The Vigenère cipher uses a keyword to perform multiple Caesar shifts.

For example:

Plaintext: ATTACK

Key: LEMONL

Each character is encrypted using a different shift based on the corresponding key character.

Formula

$$C_i = (P_i + K_i) \bmod 26$$

Characteristics

- Key usage: Repeating keyword.
- Complexity: Moderate.
- Security strength: Stronger than simple substitution ciphers.
- Uses multiple substitution alphabets.

Advantage

Because different plaintext letters can produce different ciphertext letters depending on the key position, simple frequency analysis becomes more difficult.

Weakness

If the key is short and repeated, techniques such as **Kasiski examination** and statistical analysis can help attack it.

6. Hill Cipher

The Hill cipher uses **matrix multiplication and modular arithmetic**.

Plaintext characters are converted into numerical values.

For example:

A = 0

B = 1

C = 2

...

Z = 25

A plaintext block is represented as a vector.

Encryption

$$C = KP \bmod 26$$

Where:

- C = ciphertext vector
- K = encryption matrix
- P = plaintext vector

Decryption

$$P = K^{-1}C \bmod 26$$

The key matrix must have a valid modular inverse.

Characteristics

- Key usage: Matrix.
 - Complexity: Higher.
 - Security strength: Stronger than basic Caesar and monoalphabetic substitution.
 - Uses linear algebra.
 - Uses modular arithmetic.
-

7. Comparative Analysis

Feature	Caesar	Monoalphabetic	Vigenère	Hill
Key	Shift value	Substitution alphabet	Keyword	Matrix
Number of alphabets	1	1	Multiple	Matrix-based
Complexity	Very low	Low	Medium	High
Frequency analysis resistance	Very poor	Poor	Better	Better
Brute-force resistance	Very poor	Better	Better	Better
Mathematical operations	Addition	Substitution	Modular addition	Matrix multiplication
Key management	Very simple	Moderate	Moderate	Complex
Educational value	Basic	Basic	Intermediate	Advanced
Modern security	Not suitable	Not suitable	Not suitable	Not suitable

Important: Although Hill and Vigenère are stronger than Caesar and monoalphabetic ciphers in the classical-cipher context, none of these classical schemes should be used as a modern secure encryption system.

8. Security Ranking

A reasonable classical-cipher ranking is:

Caesar



Monoalphabetic



Vigenère



Hill

In terms of resistance to basic classical attacks, **Vigenère and Hill are substantially stronger than Caesar and monoalphabetic substitution**, but their security depends on the key and attack model.

9. Why Caesar Is Weak

Caesar has only a small number of possible keys.

For example:

A → D

B → E

C → F

...

An attacker can test every possible shift.

Therefore:

Small Key Space

↓

Easy Brute Force

↓

Easy Decryption

↓

Low Security

10. Why Monoalphabetic Is Vulnerable

Although it has many possible substitution alphabets, each plaintext letter always maps to the same ciphertext letter.

Therefore, the statistical characteristics of the language remain visible.

Plaintext Frequency

↓

Ciphertext Frequency

↓

Frequency Analysis

↓
Possible Key Recovery

11. Why Vigenère Is Stronger

Vigenère uses multiple substitution alphabets.

Same plaintext letter

|
+---- Position 1 → Cipher A
|
+---- Position 2 → Cipher F
|
+---- Position 3 → Cipher M

Therefore, simple frequency analysis is less effective.

However, repeated keys can introduce patterns.

12. Why Hill Is More Complex

Hill cipher encrypts blocks of characters rather than treating every character independently.

Plaintext

|

v

Convert to Numbers

|

v

Create Vector

|

v

Multiply by Key Matrix

|

v

Modulo 26

|

v

Ciphertext

This provides diffusion across characters within a block.

13. Final Analysis

Among the four classical techniques:

- **Caesar** → simplest and weakest.
- **Monoalphabetic** → stronger key space but vulnerable to frequency analysis.
- **Vigenère** → stronger because it uses multiple substitution alphabets.
- **Hill** → mathematically more complex and provides block-level transformation.

Conclusion

For the classical techniques considered, **Hill and Vigenère provide greater resistance to basic attacks than Caesar and monoalphabetic substitution**. Hill is particularly interesting because it combines substitution with matrix operations and modular arithmetic.

However, these are historical/classical encryption techniques and **should not be considered secure against modern cryptanalysis**. Modern applications should use standardized algorithms such as AES or authenticated-encryption schemes.

QUESTION 5 – CONCEPT MAP

Number Theory Concepts Used in Cryptography

Topic

Modular Inverse, Extended Euclidean Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem

1. Introduction

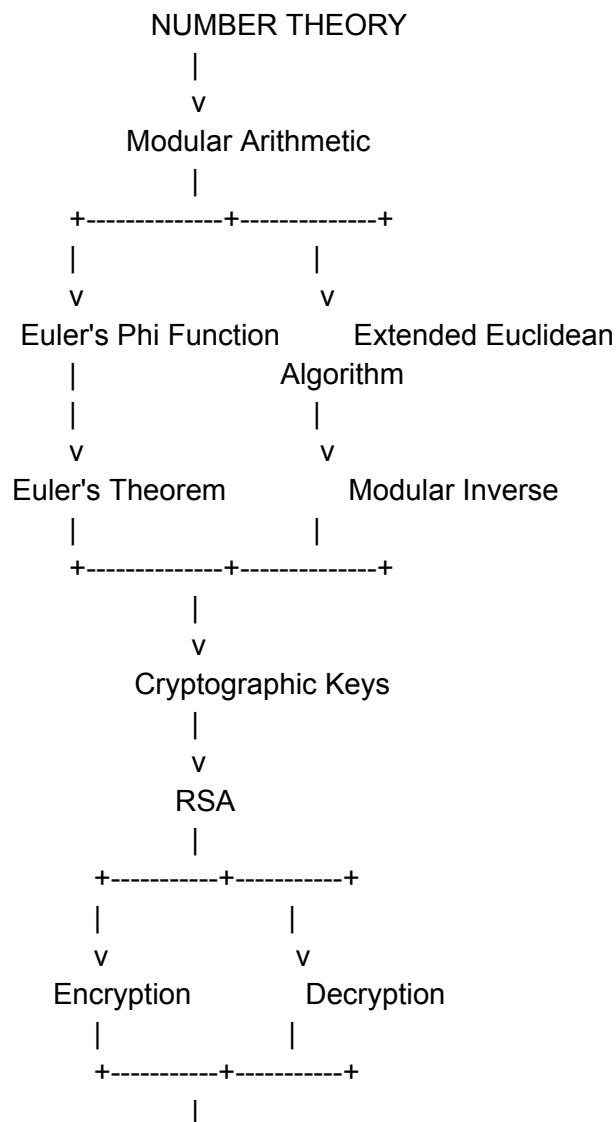
Number theory provides the mathematical foundation for many cryptographic algorithms.

Important concepts include:

- Modular arithmetic
- Modular inverse
- Extended Euclidean Algorithm
- Fermat's Little Theorem
- Euler's Phi Function
- Euler's Theorem

These concepts are particularly important in public-key cryptography such as **RSA**.

2. Main Concept Map



3. Modular Arithmetic

Modular arithmetic deals with remainders.

For example:

$$17 \bmod 5 = 2$$

because:

$$17 \div 5 = 3 \text{ remainder } 2$$

Cryptographic algorithms use modular arithmetic to keep calculations within a defined numerical range.

4. Modular Inverse

The modular inverse of a modulo m is a value x satisfying:

$$ax \equiv 1 \pmod{m}$$

Example

Find the inverse of 3 modulo 7.

We need:

$$3x \equiv 1 \pmod{7}$$

Try:

$$3 \times 5 = 15$$

and:

$$15 \bmod 7 = 1$$

Therefore:

$$3^{-1} \equiv 5 \pmod{7}$$

So the modular inverse is **5**.

5. Extended Euclidean Algorithm

The Extended Euclidean Algorithm is used to find:

$$\gcd(a,b)$$

and coefficients x,y such that:

$$ax+by=\gcd(a,b)$$

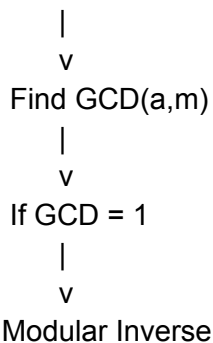
If:

$$\gcd(a,m)=1$$

the algorithm can be used to find the modular inverse of a modulo m .

Relationship

Extended Euclidean Algorithm



6. Euler's Phi Function

Euler's Phi function, written as:

$$\phi(n)$$

counts the positive integers less than or equal to n that are relatively prime to n .

For a prime number p :

$$\phi(p)=p-1$$

For two distinct primes p and q :

$$\phi(pq)=(p-1)(q-1)$$

This relationship is fundamental to RSA.

7. Example of Euler's Phi Function

Let:

$$n=15$$

Numbers relatively prime to 15 are:

1, 2, 4, 7, 8, 11, 13, 14

Therefore:

$$\phi(15)=8$$

8. Euler's Theorem

Euler's theorem states that if:

$$\gcd(a,n)=1$$

then:

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

This theorem is important in understanding why RSA encryption and decryption can reverse each other under the required mathematical conditions.

9. Fermat's Little Theorem

Fermat's Little Theorem states that for a prime p , if:

$$p \nmid a$$

then:

$$a^{p-1} \equiv 1 \pmod{p}$$

It is a special case of Euler's theorem because:

$$\phi(p) = p-1$$

Therefore:

Fermat's Little Theorem

↓

Special case of

↓

Euler's Theorem

10. Relationship Between Euler and Fermat

Euler's Theorem

|

∨

$$\gcd(a, n) = 1$$

|

∨

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

|

$$\text{If } n = p$$

|

∨

$$\phi(p) = p - 1$$

|

∨

$$a^{(p-1)} \equiv 1 \pmod{p}$$

|
v

Fermat's Little Theorem

Thus, Fermat's Little Theorem can be viewed as a special case of Euler's theorem for prime moduli.

11. Relationship With RSA

RSA uses two large prime numbers:

p, q

Then:

$n=pq$

The Euler Phi function is calculated:

$\phi(n)=(p-1)(q-1)$

A public exponent e is selected such that:

$\gcd(e, \phi(n))=1$

The Extended Euclidean Algorithm is then used to calculate the private exponent d :

$ed \equiv 1 \pmod{\phi(n)}$

Therefore:

Choose p and q

|
v

Calculate $n = pq$

|
v

Calculate $\phi(n)$

|
v

Choose e

|
v

Extended Euclidean Algorithm

|

v

Calculate d

|

v

RSA Key Pair

12. RSA Encryption

The public key consists of:

(e,n)

The plaintext is represented as M.

Encryption:

$C = M^e \bmod n$

where C is the ciphertext.

Plaintext M

|

v

Public Key (e,n)

|

v

$C = M^e \bmod n$

|

v

Ciphertext C

13. RSA Decryption

The private key consists of:

(d,n)

Decryption:

$M = C^d \bmod n$
 Ciphertext C
 ↓
 Private Key (d,n)
 ↓
 $M = C^d \bmod n$
 ↓
 Original Plaintext

14. Mathematical Connection in RSA

The important relationship is:

$$ed \equiv 1 \pmod{\phi(n)}$$

This means:

$$ed = 1 + k\phi(n)$$

for some integer k.

Euler's theorem helps establish the mathematical basis for reversing the exponentiation under the relevant conditions.

Therefore:

Euler Phi Function

↓
 $\phi(n)$
 ↓
 Choose Public Exponent e
 ↓
 Extended Euclidean Algorithm
 ↓
 Calculate Private Exponent d
 ↓

15. Role of Each Concept

Concept	Main Role in Cryptography
Modular Arithmetic	Performs calculations within a finite modulus
Modular Inverse	Finds multiplicative inverse modulo n
Extended Euclidean Algorithm	Efficiently computes GCD and modular inverses
Euler's Phi Function	Used in RSA key generation
Euler's Theorem	Provides mathematical basis for modular exponentiation relationships
Fermat's Little Theorem	Special case useful for prime modulus calculations

16. Which Concept Is Most Critical?

There is **no single concept that independently provides RSA security**. RSA depends on several interconnected mathematical concepts.

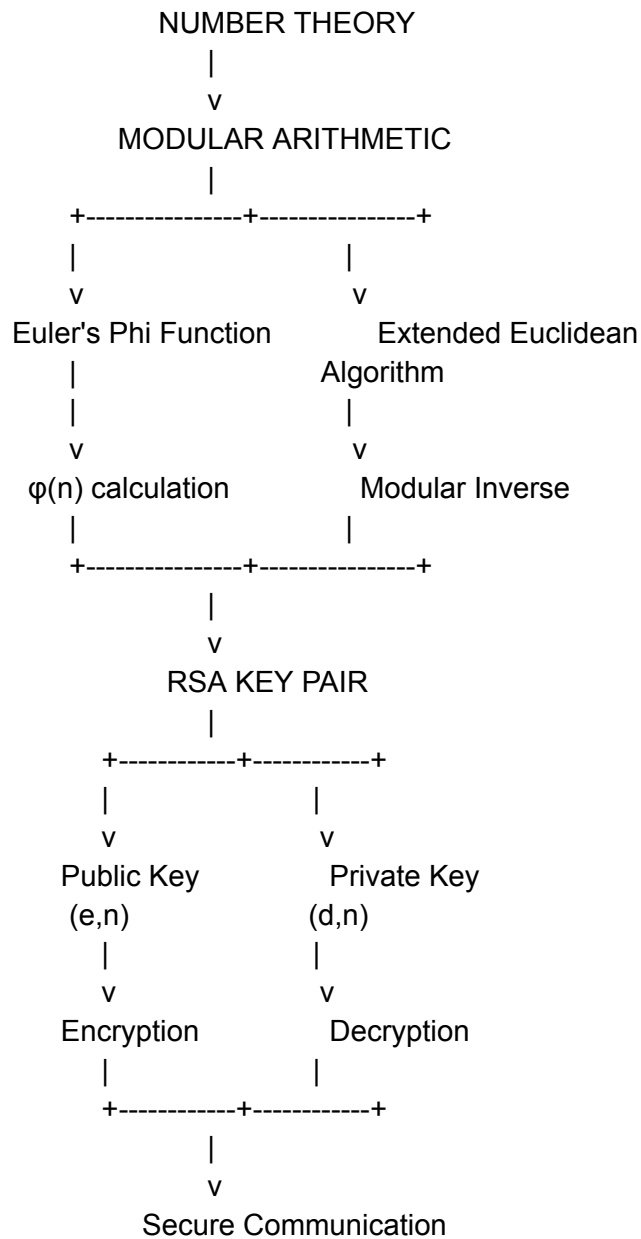
However, for **practical RSA key generation**, the **modular inverse / Extended Euclidean Algorithm relationship is particularly critical** because the private exponent d must satisfy:

$$ed \equiv 1 \pmod{\phi(n)}$$

Without obtaining the correct modular inverse, the RSA private key cannot be correctly generated.

At the same time, **Euler's Phi function is fundamental to constructing the RSA key relationship**, so the concepts should be understood together.

17. Integrated Concept Map



18. Comparative Summary

Concept	Mathematical Purpose	Cryptographic Importance
Modular Inverse	Finds inverse under a modulus	Essential in RSA key generation
Extended Euclid	Efficient inverse calculation	Used to calculate RSA private exponent

Euler Phi	Counts coprime integers	Fundamental to RSA
Euler's Theorem	Establishes modular exponentiation relationship	Supports RSA mathematical correctness
Fermat's Theorem	Prime-modulus special case	Useful in number-theoretic cryptography

19. Real-World Application

These mathematical concepts are used in public-key cryptography systems.

Examples include:

- RSA encryption.
- RSA digital signatures.
- Cryptographic key generation.
- Public-key infrastructure.
- Secure communication protocols.

The mathematical concepts allow computers to perform cryptographic operations efficiently while making certain inverse problems difficult for attackers when appropriately large parameters and secure implementations are used.

20. Conclusion

Number theory forms an essential mathematical foundation for public-key cryptography.

The **Euler Phi Function** helps establish the RSA modulus relationship, while the **Extended Euclidean Algorithm** calculates the modular inverse needed to obtain the private exponent. **Euler's Theorem** explains important modular exponentiation properties, and **Fermat's Little Theorem** provides a related result for prime numbers.